Internal Energy, Specific Heat and Correlation Function of the Bond-Random Ising Model

Report on H. Nishimori, *Prog. Theor. Phys.* **66**, 1169 (1981)

EE22B003 — Advanced Statistical Physics

1. Introduction

The paper studies the Ising model on a lattice whose nearest-neighbour couplings are random in sign (the $\pm J$ model). Each bond carries $T_{ij} = \pm 1$, taking $+1$ with probability p and -1 with probability $1-p$. Writing this as a two-level distribution,

$$P(T_{ij}) = \frac{e^{K_p T_{ij}}}{2\cosh K_p}, \qquad e^{-2K_p} = \frac{1-p}{p}, \tag{1}$$

the partition function for a fixed bond configuration is $Z(\{T\}, K) = \sum_{\{\sigma\}} \exp\bigl(K \sum_{\langle ij \rangle} T_{ij}\sigma_i\sigma_j\bigr)$, with $K = \beta J$. The curve $K = K_p$ in the (p, T) plane is the *Nishimori line*, on which exact results can be obtained despite the disorder. The limiting cases are familiar. At $p = 1$ the model is the pure Ising ferromagnet, solved by Onsager, with $\tanh K_c = 1/(\sqrt{2}+1)$ in two dimensions; at $p = 0$ it is purely antiferromagnetic, and for $p < \frac{1}{2}$ the system is an exact replica of the $p > \frac{1}{2}$ one with ferro and antiferro bonds exchanged. At intermediate p the competition between the two bond types produces *frustration*: a plaquette whose bonds multiply to -1 cannot satisfy all its bonds at once, so at $T = 0$ minimising the energy has exponentially many solutions. The question is how this shapes the phase diagram (Fig. 1), whose paramagnetic, ferromagnetic and (in $d > 2$) spin-glass regions meet at a multicritical point on the Nishimori line.

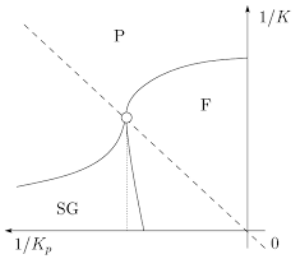


Figure 1: Phase diagram in the (p, T) plane: spin glass, paramagnetic and ferromagnetic phases, the Nishimori line N, and the multicritical point at p_c.

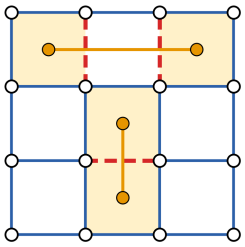


Figure 3: Frustrated plaquettes on a square lattice (red bonds are antiferromagnetic). They occur in pairs, and the transition is reached when pairs separate to infinity.

Spin glass and frustration

Each frustrated plaquette admits four equally good ground states (one unsatisfied bond among four choices). If a fraction $(1-p)$ of the N plaquettes is frustrated, there are of order $4^{(1-p)N}$ minima, exponentially many in N; such a phase is called a spin glass. In two dimensions it can exist only at $T = 0$: when antibonds percolate, a domain wall can follow an antibond chain of any length L at an energy cost of only $O(1)$, so by Peierls' argument no ordered or glassy phase survives at finite temperature. In $d \geq 3$ antibonds percolate only along lines, not sheets, a domain wall costs $O(L^2)$, and the argument gives a spin-glass phase that persists at finite temperature.

2. Method

2.1 Frustration and its entropy

Let W be the product of the T_{ij} around each plaquette; this configuration of W is the "frustration". The gauge transformation

$$T_{ij} \to T_{ij} S_i S_j, \qquad \sigma_i \to \sigma_i S_i \tag{2}$$

by any field $S_i = \pm 1$ on the lattice sites leaves W unchanged, because $\sigma'_i\sigma'_j = (\sigma_i S_i)(\sigma_j S_j) = (\sigma_i\sigma_j)(S_i S_j)$. It also leaves Z unchanged, so Z depends on the bonds only through W. Starting from the product measure $P(\tau) = \prod_{\langle ij \rangle} e^{K_p \tau_{ij}}/(2\cosh K_p)$, the probability of a given frustration pattern is the sum over all bond configurations $\tau^{(0)}(W)\,\sigma\sigma$ that realise it,

$$P(W) = \frac{1}{(2\cosh K_p)^{N_b}} \sum_{\{\sigma\}} \exp\Bigl(K_p \sum_{\langle ij \rangle} \tau^{(0)}_{ij}(W)\sigma_i\sigma_j\Bigr), \tag{3}$$

where N_b is the number of bonds. This is the partition function of an ordinary Ising model. Frustrated plaquettes come in pairs (Fig. 3), and the phase transition corresponds to the separation within a pair growing without bound.

2.2 Free energy on the Nishimori line

Averaging $\ln Z$ over the bond distribution, and using gauge invariance to average over all $\{T_{ij}S_iS_j\}$ that give the same W, gives

$$-\beta[F] = \frac{1}{2^N} \sum_{\{S_i\}} \sum_{\{T_{ij}\}} P(\{T_{ij}S_iS_j\}) \ln Z(\{T_{ij}S_iS_j\}, K). \tag{4}$$

Here $\sum_{\{S\}} P(\{T_{ij}S_iS_j\}) = (2\cosh K_p)^{-N_b} \sum_{\{S\}} \exp(K_p \sum T_{ij}S_iS_j) = (2\cosh K_p)^{-N_b} Z(\{T\}, K_p)$, and therefore

$$-\beta[F] = \frac{1}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T\}} Z(\{T\}, K_p)\ \ln Z(\{T\}, K). \tag{5}$$

Moments of the energy and of the correlations follow by differentiating with respect to K and then setting $K = K_p$.

3. Results

3.1 Internal energy

With $E = -J \sum T_{ij}\sigma_i\sigma_j$ we have $\langle E\rangle = -J\,\partial_K \ln Z$, so by Eq. (5)

$$[E] = -\frac{J}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T\}} \left.\frac{\partial Z(\{T\}, K)}{\partial K}\right|_{K=K_p}. \tag{6}$$

The sum over bonds factorises, $\sum_{\{T\}} Z(\{T\}, K) = \sum_{\{\sigma\}} \prod_{\langle ij\rangle} \sum_{T_{ij}=\pm 1} e^{KT_{ij}\sigma_i\sigma_j} = 2^N (2\cosh K)^{N_b}$, and differentiating gives $2^N N_b (2\cosh K)^{N_b} \tanh K$. The result is exact on any lattice:

$$\boxed{[E] = -N_b J \tanh K_p} \tag{7}$$

3.2 Specific heat

The same steps applied to the second moment give $[\langle E^2\rangle] = \frac{J^2}{2^N (2\cosh K_p)^{N_b}}\, \partial_K^2 \big[2^N (2\cosh K)^{N_b}\big]_{K=K_p}$, that is

$$[\langle E^2\rangle] = J^2 \big(N_b^2 \tanh^2 K_p + N_b\ \mathrm{sech}^2 K_p\big). \tag{8}$$

Since $[X^2] \geq [X]^2$, the configurationally averaged specific heat obeys

$$[C] \leq \frac{K^2}{J^2} \Big([\langle E^2\rangle] - [E]^2\Big) = \frac{K_p^2}{J^2}\, N_b J^2\, \mathrm{sech}^2 K_p = N_b\, K_p^2\, \mathrm{sech}^2 K_p, \tag{9}$$

which is finite and smooth along the Nishimori line, including at the multicritical point.

3.3 Correlation functions

The thermal correlation $\langle S_iS_j\rangle_K = Z^{-1} \sum_{\{S\}} S_iS_j e^{K \sum T_{lm}S_l\S_m}$ transforms under the gauge transformation as

$$\langle S_iS_j\rangle_K(\{T_{lm}S_lS_m\}) = S_iS_j\ \langle S_iS_j\rangle_K(\{T_{lm}\}). \tag{10}$$

Combining this with the identity $\sum_{\{S\}} S_iS_j \exp(K_p \sum T_{lm}S_lS_m) = Z(\{T\}, K_p)\langle S_iS_j\rangle_{K_p}$, the average becomes a sum over all weights $P(T)$ and

$$[\langle S_iS_j\rangle_K] = \big[\langle S_iS_j\rangle_{K_p}\ \langle S_iS_j\rangle_K\big]. \tag{11}$$

Because every thermal correlation satisfies $|\langle S_iS_j\rangle_K| \leq 1$,

$$\big|[\langle S_iS_j\rangle_K]\big| \leq [\langle S_iS_j\rangle_{K_p}]. \tag{12}$$

4. Summary

Gauge invariance makes the free energy, energy, specific heat and correlations of the $\pm J$ Ising model exactly computable on the Nishimori line $K = K_p$.

- **Energy:** $[E] = -N_b J \tanh K_p$ holds exactly in any dimension.
- **Specific heat:** $[C] \leq N_b K_p^2\, \mathrm{sech}^2 K_p$ is finite on the line, so the multicritical point has no divergent heat capacity: the transition is of a percolation type rather than a "boiling" (latent-heat) type.
- **Phase boundary:** by Eq. (12), a system that is paramagnetic on the Nishimori line (correlations decaying to zero at large $|i-j|$) is paramagnetic at all temperatures for that p, so the boundary is vertical or curves inward below the multicritical point.
- **No spin glass on the line:** at $K = K_p$, $m = q = [\langle S_iS_j\rangle^2]$, whereas a spin glass needs $q > 0$ with $m = 0$. This is impossible on the line and at higher temperature. For $T < T_c$, $p < p_c$ ferromagnetism is ruled out, but for $d > 2$ a spin-glass phase can exist.

References

[1] H. Nishimori, "Internal Energy, Specific Heat and Correlation Function of the Bond-Random Ising Model," *Prog. Theor. Phys.* **66**, 1169 (1981).